

MACHINE-CHECKED CANONICAL HEIGHTS AND RANK LOWER BOUNDS, UNIFORMLY IN THE CURVE

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ABSTRACT. We report a machine-checked construction of the canonical (Néron–Tate) height, and of rank lower bounds, for elliptic curves over $\mathbb{Q}$, which is *uniform in the curve*. Working in Lean 4 over `mathlib`: for every Weierstrass curve $W/\mathbb{Q}$ whose b -invariants are integers, whose discriminant Δ is nonzero, and which has no rational 2-torsion, we construct $\widehat{h}: W(\mathbb{Q}) \rightarrow \mathbb{R}$, prove the parallelogram law for it *exactly*, deduce $\widehat{h}(kR) = k^2\widehat{h}(R)$ and bi-additivity of the associated pairing, and conclude: if n rational points have nonzero Gram determinant, then $n \leq \text{rank}_{\mathbb{Z}} W(\mathbb{Q})$. Every constant is an explicit polynomial in the coefficients of the curve. The predecessor of this paper proved $\text{rank} \geq 2$ for `389a1` and $\text{rank} \geq 3$ for `5077a1` through certificate suites built by hand for each curve; here both are re-derived by instantiating the uniform machine, with the earlier interval arithmetic as the only per-curve numerical input.

The mathematical content is a reduction. The secant forms that control $x(P+Q)$ and $x(P-Q)$ restrict on the diagonal to the duplication pair (Φ, Ψ) , so the secant content problem reduces to the resultant identity $\text{Res}(\Phi, \Psi) = \Delta^2$ rather than needing an elimination theory of its own; twin Bézout certificates in the two monomial weights, together with a cubing of the duplication identity, give a divisor bound Δ^6 for the content of the secant triple, on every curve. Exactly one hypothesis survives universalization — no rational 2-torsion — and it is necessary, because Ψ vanishes precisely at rational 2-torsion. The price of uniformity is paid in the size of the constants, which we tabulate. Kernel axioms are `[propext, Classical.choice, Quot.sound]` throughout, with no `sorry`, no `native_decide` and no project-level axiom. These are rank *lower* bounds; nothing here bears on descent, on L -functions, or on the Birch–Swinnerton-Dyer conjecture.

1. INTRODUCTION

1.1. What is new. In [2] the canonical height of Néron and Tate was constructed inside a proof assistant for the first time, and used to prove $2 \leq \text{rank } E(\mathbb{Q})$ for $E = \text{389a1}$ and $3 \leq \text{rank } E'(\mathbb{Q})$ for $E' = \text{5077a1}$. Two components of that work were already curve-independent: the passage from a doubling window to a canonical height, and the passage from a nonvanishing Gram determinant to a rank bound. Everything between them — the duplication estimates, the secant estimates, the quasi-parallelogram law and hence bi-additivity — was written out by hand, once per curve, in some forty files.

This paper reports that the whole of the middle is now curve-independent. The main theorem is the following.

Theorem 1.1 (`rank_ge_universal`). *Let W be a Weierstrass curve over $\mathbb{Q}$ (`mathlib`’s `WeierstrassCurve.Affine` $\mathbb{Q}$), and suppose*

- (i) *the b -invariants of W are integers, $b_i = B_i$ with $B_2, B_4, B_6, B_8 \in \mathbb{Z}$;*
- (ii) *$\Delta := -B_2^2 B_8 + 9B_2 B_4 B_6 - 8B_4^3 - 27B_6^2 \neq 0$;*
- (iii) *$W(\mathbb{Q})$ has no rational 2-torsion: $2R = O$ implies $R = O$.*

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Let $\hat{h}$ be the canonical height of §5 and $\langle \cdot, \cdot \rangle$ the pairing of §6. If $P: \{1, \dots, n\} \rightarrow W(\mathbb{Q})$ satisfies

$$\det(\langle P_i, P_j \rangle)_{i,j} \neq 0,$$

then $n \leq \text{rank}_{\mathbb{Z}} W(\mathbb{Q})$.

We call a curve satisfying (i)–(iii) *admissible*. Nothing else is assumed: no reduced model, no minimality, no bound on the conductor, no knowledge of the torsion subgroup beyond (iii). All of the constants that appear in the construction are explicit polynomials in $|B_2|, |B_4|, |B_6|, |B_8|$.

Two specializations are proved in the same file, because they are the shapes one actually checks. At $n = 2$ the Gram determinant is the classical regulator expression,

$$\det(\langle P_i, P_j \rangle) = \hat{h}(P) \hat{h}(Q) - \langle P, Q \rangle^2 \quad (\text{gram_det_fin_two}), \quad (1)$$

so a nonzero regulator forces $\text{rank} \geq 2$ (`rank_ge_two_universal`); at $n = 3$ the corresponding expansion is `gram_det_fin_three`.

Theorem 1.1 rests on the following, which is where the mathematical work is.

Theorem 1.2 (`canheight_parallelogram_universal`, `par`, `mul`, `biAdditive_pairingW`). *Let W be admissible. There is a function $\hat{h}: W(\mathbb{Q}) \rightarrow \mathbb{R}$, defined as*

$$\hat{h}(R) = \lim_{n \rightarrow \infty} \frac{\log H(x(2^n R))}{4^n},$$

such that, formally:

- (1) $\hat{h} \geq 0$, $\hat{h}(O) = 0$ and $\hat{h}(-R) = \hat{h}(R)$;
- (2) $|\hat{h}(R) - \log H(x(R))| \leq \frac{1}{3} \log \kappa$, where κ is the explicit constant of §3;
- (3) $\hat{h}(P + Q) + \hat{h}(P - Q) = 2\hat{h}(P) + 2\hat{h}(Q)$ for all $P, Q \in W(\mathbb{Q})$, with no side conditions;
- (4) $\hat{h}(kR) = k^2 \hat{h}(R)$ for all $k \in \mathbb{Z}$ and all R ;
- (5) the pairing $\langle P, Q \rangle := \frac{1}{2}(\hat{h}(P+Q) - \hat{h}(P) - \hat{h}(Q))$ is symmetric and additive in each variable, and $\langle P, P \rangle = \hat{h}(P)$.

Here $H(a/b) = \max(|a|, b)$ for a reduced fraction, and $x(R)$ is the x -coordinate of R , with $x(O) := 0$ (harmless, since $H(0) = 1$). Every statement in Theorems 1.1 and 1.2 is a Lean 4 theorem; `#print axioms` on each reports a subset of `[propext, Classical.choice, Quot.sound]` and nothing else. §11 gives the theorem-by-theorem map to the sources and the verification data; §12 states, specifically, what is *not* proved — these are rank lower bounds, and nothing here concerns descent or L -functions.

Note the strengthening in Theorem 1.2(3) and (4). In [2] the parallelogram law carried four non-torsion side conditions, discharged for the two worked curves by proving their torsion subgroups trivial. Here it is unconditional on any admissible curve: the degenerate configurations are closed separately, and the one place where genuine torsion could intrude — 2-torsion — is excluded by hypothesis (iii). That is what makes the multiple law and bi-additivity come out without bookkeeping.

1.2. Relation to the predecessor, and what the two curves are for. The results of [2] are re-derived here, through the uniform machine:

$$2 \leq \text{rank}_{\mathbb{Z}} E(\mathbb{Q}), \quad E = 389a1: y^2 + y = x^3 + x^2 - 2x \quad (\text{E389a1_rank_ge_two_universal}), \quad (2)$$

$$3 \leq \text{rank}_{\mathbb{Z}} E'(\mathbb{Q}), \quad E' = 5077a1: y^2 + y = x^3 - 7x + 6 \quad (\text{E5077a1_rank_ge_three_universal}). \quad (3)$$

These are not new arithmetic facts, and they were not new in [2] either: the ranks of these two curves are classical. Their role here is different. They are the evidence that the uniform machine is not vacuous and that it reproduces, rather than merely parallels, the hand-built results: (2) and (3) are obtained by instantiating Theorem 1.1 at the two hypothesis packages and feeding it the

certified interval arithmetic of [2] — the regulator positivity computations — as the sole numerical input. No height estimate specific to either curve appears anywhere in the chain. §8 describes the transfer and records a verification-methodology point that we think is of independent interest: the uniform canonical height and the per-curve one turn out to be equal *by rfl*.

1.3. The obstruction, and how it is avoided. The reason the middle of [2] was per-curve is easy to state. Both halves of the construction — doubling and addition — need two things about an explicit pair of integer forms: a bound on their *content* (the common factor that cancels when the fraction is reduced), and two-sided bounds on their *size*. Per curve, the content is computed by an extended Euclidean algorithm on the actual integer polynomials, and the size by summing actual coefficients. Both outputs are numbers, not formulas, and both computations must be redone from scratch for each new curve.

The duplication half was made uniform by a resultant identity: writing $x = X/Y$ and homogenizing,

$$\Phi(X, Y) = X^4 - b_4 X^2 Y^2 - 2b_6 X Y^3 - b_8 Y^4, \quad \Psi(X, Y) = 4X^3 Y + b_2 X^2 Y^2 + 2b_4 X Y^3 + b_6 Y^4, \quad (4)$$

one has $\text{Res}(\Phi, \Psi) = \Delta^2$ as an identity in $\mathbb{Z}[b_2, b_4, b_6, b_8]$ modulo the relation $4b_8 = b_2 b_6 - b_4^2$, and the Sylvester cofactors give two explicit Bézout witnesses. That was already in place, and is summarized in §3. The secant half looked as though it would need its own elimination theory — a resultant in two sets of variables, on forms twice as large. It does not, and the reason is the first of four ideas this paper contributes.

First: diagonal degeneration. Write $x_i = u_i/v_i$ in lowest terms. The three integer forms controlling addition are $\mathcal{D} = (u_1 v_2 - u_2 v_1)^2$ and the bidegree-(2, 2) forms $\mathcal{S}, \mathcal{P}$ of (8) below, which satisfy $\mathcal{D} \cdot (x(P+Q) + x(P-Q)) = \mathcal{S}$ and $\mathcal{D} \cdot x(P+Q)x(P-Q) = \mathcal{P}$. Restricted to the diagonal $(u_2, v_2) = (u_1, v_1)$ they are the duplication pair:

$$\mathcal{S}(u, v, u, v) = \Psi(u, v), \quad \mathcal{P}(u, v, u, v) = \Phi(u, v), \quad \mathcal{D}(u, v, u, v) = 0. \quad (5)$$

The secant construction degenerates, honestly, onto the duplication construction. Consequently the secant content problem does not need a new keystone: eliminating the second point against $\mathcal{D}$ lands on Φ and Ψ at the first point, and $\text{Res}(\Phi, \Psi) = \Delta^2$ finishes the job.

Second: twin certificates. Elimination produces the duplication forms *squared*, weighted by a monomial in the second point's coordinates. Sylvester's matrix supplies the resultant in both monomial weights, and both are needed: one gets

$$\sigma_1 \mathcal{D} + \sigma_2 \mathcal{S} = v_2^3 \Psi_1^2 \quad \text{and} \quad \sigma'_1 \mathcal{D} + \sigma'_2 \mathcal{S} = u_2^3 \Psi_1^2,$$

with explicit integer cofactors, and similarly for $\mathcal{P}$ and Φ_1 . Neither weight alone is usable — u_2 and v_2 can each be arbitrarily divisible — but since $\gcd(u_2, v_2) = 1$, so are u_2^3 and v_2^3 , and the two weights cancel each other. A common divisor of the secant triple therefore divides Ψ_1^2 and Φ_1^2 outright.

Third: the squaring trick. What is now available are the *squares* Φ_1^2, Ψ_1^2 , while the duplication Bézout identity is linear, $F_1 \Phi + G_1 \Psi = \Delta^2 X^7$. Cubing it puts every term into the ideal generated by Φ^2 and Ψ^2 :

$$(F\Phi + G\Psi)^3 = (F^3\Phi + 3F^2G\Psi)\Phi^2 + (3FG^2\Phi + G^3\Psi)\Psi^2. \quad (6)$$

Hence a common divisor of the secant triple divides $(\Delta^2 X^7)^3 = \Delta^6 X^{21}$ and likewise $\Delta^6 Y^{21}$, and coprimality of X, Y leaves Δ^6 . That is exactly where the exponent 6 comes from; it is the price of the cube, and we do not claim it is optimal.

Fourth: the honest residue. Per curve, [2] discharged all non-torsion side conditions from the triviality of the torsion subgroup, proved by enumeration. Uniformly, exactly one hypothesis survives, and it is hypothesis (iii): no rational 2-torsion. It is not an artifact. On the curve one has $\Psi(x) = (2y + a_1 x + a_3)^2$ (`psiX_eq_sq`), so Ψ vanishes at $x(R)$ precisely when $2R = O$; at such a point the duplication map has no denominator, the height window's construction has nothing to

say, and the whole chain stops. Curves with rational 2-torsion are genuinely outside the scope of Theorem 1.1; they would need a per-step variant, which we have not carried out.

1.4. Related formalization. `mathlib` [5] provides the Weierstrass curve type, the group law on the affine, projective and Jacobian models with a full associativity proof, variable change, normal forms, division polynomials, and reduction over discrete valuation rings. At the commit pinned for this work (§11) it contains no arithmetic height theory, no Mordell–Weil theorem, and no descent; we checked this directly against the pinned copy.

On `mathlib master` at the time of writing there is a `Mathlib/NumberTheory/Height/` directory — `Basic`, `Northcott`, `MvPolynomial`, `NumberField`, `Projectivization`, `EllipticCurve` — developing heights over fields with a theory of absolute values, and a `Mathlib/GroupTheory/Descent.lean` of M. Stoll giving an abstract descent theorem (a height function with suitable growth relative to an endomorphism forces finite generation), one of the ingredients of Mordell–Weil. The elliptic-curve height file states as its goal the approximate parallelogram law $|h(P+Q) + h(P-Q) - 2(h(P) + h(Q))| \leq C$ for the naïve height, lists it as a TODO, and currently contains one theorem: that the sum-and-difference map on $\mathbb{P}^2$ induced by a Weierstrass curve is a morphism, in the form $\exists C \forall x, |\log H(F(x)) - 2 \log H(x)| \leq C$. That is the same estimate our §4 proves, in far greater generality of ground field — and with a non-explicit constant. The difference matters for our purposes: an unquantified C cannot be fed to interval arithmetic, whereas the constants below are polynomials in the coefficients and can be evaluated. Neither the canonical height nor any rank bound is present there. Our development is independent of both files.

Buzzard’s FLT project [1] targets a modularity lifting theorem sufficient for Fermat, not the full modularity theorem. Gross–Zagier, Kolyvagin and Heegner point theory are, to our knowledge, unformalized anywhere.

2. SETUP AND THE b -INVARIANT BASIS

Throughout, W is a Weierstrass curve over $\mathbb{Q}$ with coefficients $(a_1, a_2, a_3, a_4, a_6)$, and $W(\mathbb{Q})$ is `mathlib`’s `WeierstrassCurve.Affine.Point`: the inductive type with constructors `zero` and `some`, the latter carrying a nonsingular affine rational point, with `mathlib`’s `AddCommGroup` instance. The group law requires only that the ground field have decidable equality; nonsingularity of the curve is not an instance hypothesis, which is why $\Delta \neq 0$ appears explicitly in Theorem 1.1.

The b -invariants are

$$b_2 = a_1^2 + 4a_2, \quad b_4 = 2a_4 + a_1a_3, \quad b_6 = a_3^2 + 4a_6, \quad b_8 = a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2,$$

subject to `mathlib`’s `b_relation`

$$4b_8 = b_2b_6 - b_4^2, \tag{7}$$

and $\Delta = -b_2^2b_8 + 9b_2b_4b_6 - 8b_4^3 - 27b_6^2$. Hypothesis (i) of Theorem 1.1 says that the four b -invariants are the images of integers B_2, B_4, B_6, B_8 ; the integral form of (7) is then *derived* from `mathlib`’s rational one rather than assumed (`b_relation_int`). We drop the capitals below and write b_i for the integers.

Choosing the b -invariants as the ground basis, rather than the a_i , is not cosmetic. Every object in this paper — Φ , Ψ , Δ , and the three secant forms — is a polynomial in the b_i alone. Expressed over $\mathbb{Z}[a_1, \dots, a_6]$ the Bézout cofactors of §3 would be unmanageable; over $\mathbb{Z}[b_2, b_4, b_6, b_8]$ the four of them comprise 168 monomials in total (55, 54, 26 and 33 respectively), of total degree at most nine in $(X, Y, b_2, b_4, b_6, b_8)$, which the kernel checks by `ring` without complaint. In that free polynomial ring $\text{Res}(\Phi, \Psi)$ is not literally Δ^2 ; the two agree modulo (7), and the correction multiplier is eight monomials, so each identity closes with a single `linear_combination`.

Definition 2.1 (`naiveHeight`, `nh`, `X`). For $q \in \mathbb{Q}$ set $H(q) := \max(|\text{num}(q)|, \text{den}(q))$, using Lean’s reduced representation; $H(q) \geq 1$ always. Define $x: W(\mathbb{Q}) \rightarrow \mathbb{Q}$ by $x(O) = 0$ and $x(\text{some at } (X, Y)) = X$, and $h(R) := \log H(x(R))$.

Remark 2.2 (Notation). In the Lean sources the two points' reduced coordinates are named (a_1, b_1, a_2, b_2) , which collides typographically with the curve's a - and b -invariants. Here we write $x_i = u_i/v_i$ with $\gcd(u_i, v_i) = 1$ and reserve a_i, b_i for the invariants. The translation is $u_i \leftrightarrow a_i$, $v_i \leftrightarrow b_i$ in the arguments of `DDz`, `Sfz`, `Prz` and their certificates.

3. THE DUPLICATION SIDE

This section records what [2] already had in universal form; it is included because the secant side reduces to it.

Proposition 3.1 (`bezout_X`, `bezout_Y`, `dvd_disc_sq_of_isCoprime`). *With Φ, Ψ as in (4) there are explicit $F_1, G_1, F_2, G_2 \in \mathbb{Z}[b_2, b_4, b_6, b_8][X, Y]$, binary cubics in (X, Y) , such that, granted (7),*

$$F_1\Phi + G_1\Psi = \Delta^2 X^7, \quad F_2\Phi + G_2\Psi = \Delta^2 Y^7.$$

Consequently, if $\gcd(X, Y) = 1$, every common divisor of $\Phi(X, Y)$ and $\Psi(X, Y)$ divides Δ^2 .

Proposition 3.2 (`max_abs_le`, `disc_sq_mul_le`). *With $M = \max(|X|, |Y|)$,*

$$\max(|\Phi|, |\Psi|) \leq CU \cdot M^4, \quad \Delta^2 M^4 \leq CL \cdot \max(|\Phi|, |\Psi|),$$

where

$$CU = \max(1 + |b_4| + 2|b_6| + |b_8|, 4 + |b_2| + 2|b_4| + |b_6|)$$

is the coefficient sum and CL is the corresponding sum of absolute coefficients of the Bézout cofactors, $CL = \max(S_{F_1} + S_{G_1}, S_{F_2} + S_{G_2})$.

The lower bound is Proposition 3.1 read archimedeanly: each cofactor is a binary cubic, so $\Delta^2 M^7 \leq CL \cdot M^3 \max(|\Phi|, |\Psi|)$, and M^3 cancels. The same identity that removes the content analysis removes the archimedean one.

Proposition 3.3 (`nh_dblQ_le`, `nh_pow_le_of_disc_ne_zero`, `addX_self_eq_dblQ`). *Let $\text{dbl}(x) := \Phi(u, v)/\Psi(u, v)$ for $x = u/v$ reduced. Then dbl agrees with `mathlib`'s group law: for an affine point (x, y) with $2 \cdot (x, y) \neq O$ one has $x(2 \cdot (x, y)) = \text{dbl}(x)$. Moreover, if $\Delta \neq 0$ and $\Psi(u, v) \neq 0$,*

$$H(\text{dbl}(x)) \leq CU \cdot H(x)^4, \quad H(x)^4 \leq CL \cdot H(\text{dbl}(x)).$$

The unreduced fraction Φ/Ψ has content $g = \gcd(\Phi, \Psi)$ and $g \cdot H(\Phi/\Psi) = \max(|\Phi|, |\Psi|)$; then $g \geq 1$ gives the upper bound and $g \leq \Delta^2$ (Proposition 3.1) gives the lower one, the Δ^2 cancelling against the Δ^2 of Proposition 3.2. The duplication formula itself is Silverman [6, III.2.3(d)], and is proved here against `mathlib`'s `addX/slope` — a polynomial identity modulo the curve equation whose certificate is the single multiplier $-(b_2 + 8x)$ — rather than assumed.

Setting $\kappa := \max(CU, CL)$, Propositions 3.2 and 3.3 give the logarithmic doubling window $|h(2R) - 4h(R)| \leq \log \kappa$ away from O , and at O both sides vanish by the convention $x(O) = 0$ (`heightWindow_of_xdbl_ne_zero`). That is the sole input of the generic construction of [2]: given an abelian group G , a nonnegative $\lambda: G \rightarrow \mathbb{R}$ and $\kappa \geq 1$ with $|\lambda(R+R) - 4\lambda(R)| \leq \log \kappa$, the sequence $\lambda(2^n R)/4^n$ is Cauchy with geometric modulus and its limit $\hat{h}$ satisfies $\hat{h} \geq 0$, $\hat{h}(R+R) = 4\hat{h}(R)$, $\hat{h}(2^n R) = 4^n \hat{h}(R)$, $|\hat{h} - \lambda| \leq \log \kappa/3$ and $|\hat{h}(R) - \lambda(2^n R)/4^n| \leq \log \kappa/(3 \cdot 4^n)$.

4. THE SECANT SIDE

4.1. The universal formulas. Both $x(P+Q) + x(P-Q)$ and $x(P+Q) \cdot x(P-Q)$ are symmetric in the y -coordinates, so both eliminate them.

Proposition 4.1 (`addX_add_addX_negY`, `addX_mul_addX_negY`). *Over any field, for points $P = (x_1, y_1)$, $Q = (x_2, y_2)$ on a Weierstrass curve with $x_1 \neq x_2$,*

$$x(P+Q) + x(P-Q) = \frac{2x_1x_2(x_1 + x_2) + b_2x_1x_2 + b_4(x_1 + x_2) + b_6}{(x_1 - x_2)^2},$$

$$x(P+Q) \cdot x(P-Q) = \frac{x_1^2 x_2^2 - b_4 x_1 x_2 - b_6(x_1 + x_2) - b_8}{(x_1 - x_2)^2}.$$

Both are proved against `mathlib`'s `addX` and `slope` by `linear_combination` modulo the two curve equations. The b -invariant basis is again what makes the certificates small: the sum identity has cofactors 2, 2.

4.2. The integer forms. Homogenizing Proposition 4.1 at $x_i = u_i/v_i$ gives three integer forms of bidegree (2, 2):

$$\begin{aligned} \mathcal{D} &= (u_1 v_2 - u_2 v_1)^2, \\ \mathcal{S} &= 2u_1 u_2 (u_1 v_2 + u_2 v_1) + b_2 u_1 u_2 v_1 v_2 + b_4 (u_1 v_2 + u_2 v_1) v_1 v_2 + b_6 v_1^2 v_2^2, \\ \mathcal{P} &= u_1^2 u_2^2 - b_4 u_1 u_2 v_1 v_2 - b_6 (u_1 v_2 + u_2 v_1) v_1 v_2 - b_8 v_1^2 v_2^2. \end{aligned} \tag{8}$$

Proposition 4.2 (`DDz_spec`, `Sfz_spec`, `Prz_spec`, `sum_relation`, `prod_relation`). *The bihomogenization is exact: with $x_3 = x(P+Q)$ and $x_4 = x(P-Q)$,*

$$\mathcal{D}(x_3 + x_4) = \mathcal{S}, \quad \mathcal{D} x_3 x_4 = \mathcal{P},$$

as identities in $\mathbb{Q}$. In particular x_3 and x_4 are the two roots of the integer quadratic $\mathcal{D}T^2 - \mathcal{S}T + \mathcal{P}$.

4.3. Diagonal degeneration.

Proposition 4.3 (`Sfz_diag`, `Prz_diag`, `DDz_diag`). $\mathcal{S}(u, v, u, v) = \Psi(u, v)$, $\mathcal{P}(u, v, u, v) = \Phi(u, v)$, and $\mathcal{D}(u, v, u, v) = 0$.

Proof, as formalized. Substitute and expand: $2u^2(2uv) + b_2 u^2 v^2 + b_4(2uv)v^2 + b_6 v^4 = 4u^3 v + b_2 u^2 v^2 + 2b_4 u v^3 + b_6 v^4 = \Psi(u, v)$, and $u^4 - b_4 u^2 v^2 - b_6(2uv)v^2 - b_8 v^4 = \Phi(u, v)$. Each is one `ring` call. $\square$

Proposition 4.3 is the structural observation of this paper. It says that the secant construction, evaluated where the secant line degenerates to the tangent, returns the duplication construction — not up to a unit, not up to the relation (7), but on the nose. It is what one would hope for, and it is worth stating because of its consequence: the content problem for the secant triple can be *reduced* to the content problem for the duplication pair, and the latter has already been solved once and for all by $\text{Res}(\Phi, \Psi) = \Delta^2$. There is no second resultant to compute. The reduction is carried out in the next subsection; the certificates that carry it out are exactly the ones that produce Φ^2 and Ψ^2 from eliminating the second point, which is the x_2 -resultant statement $\text{Res}_{x_2}(\mathcal{D}, \mathcal{S}) = \psi(x_1)^2$, $\text{Res}_{x_2}(\mathcal{D}, \mathcal{P}) = \varphi(x_1)^2$ in bihomogenized form.

4.4. The content bound.

Proposition 4.4 (`psi_sq_cert`, `phi_sq_cert`, `psi_sq_certA`, `phi_sq_certA`). *There are explicit integer cofactors $\sigma_1, \sigma_2, \sigma'_1, \sigma'_2$ and $\pi_1, \pi_2, \pi'_1, \pi'_2$ in $\mathbb{Z}[b_2, b_4, b_6, b_8][u_1, v_1, u_2, v_2]$ with*

$$\begin{aligned} \sigma_1 \mathcal{D} + \sigma_2 \mathcal{S} &= v_2^3 \Psi(u_1, v_1)^2, & \sigma'_1 \mathcal{D} + \sigma'_2 \mathcal{S} &= u_2^3 \Psi(u_1, v_1)^2, \\ \pi_1 \mathcal{D} + \pi_2 \mathcal{P} &= v_2^3 \Phi(u_1, v_1)^2, & \pi'_1 \mathcal{D} + \pi'_2 \mathcal{P} &= u_2^3 \Phi(u_1, v_1)^2. \end{aligned}$$

All eight are verified by `ring`.

The pairing of weights is the point. Sylvester's matrix, cleared of denominators, delivers the resultant multiplied by a power of the leading coefficient, and which coefficient is leading depends on which variable one eliminates; carrying both eliminations gives the same target with weights u_2^3 and v_2^3 . Since $\gcd(u_2, v_2) = 1$ we have $\gcd(u_2^3, v_2^3) = 1$, and a Bézout combination cancels the weights (`dvd_of_dvd_coprime_weights`). Either certificate without its twin leaves an uncontrolled monomial factor and is useless.

Theorem 4.5 (`secant_content_universal`). *Let $b_2, b_4, b_6, b_8 \in \mathbb{Z}$ satisfy (7), and let $u_1/v_1, u_2/v_2$ be in lowest terms. Then every common divisor of $\mathcal{D}, \mathcal{S}, \mathcal{P}$ divides Δ^6 .*

Proof, as formalized. Let d divide all three. By Proposition 4.4 and weight cancellation, $d \mid \Psi(u_1, v_1)^2$ and $d \mid \Phi(u_1, v_1)^2$. Apply (6) to the Bézout identity $F_1\Phi + G_1\Psi = \Delta^2 u_1^7$ of Proposition 3.1: every term of the cube lies in the ideal generated by Φ^2 and Ψ^2 , so $d \mid (\Delta^2 u_1^7)^3 = u_1^{21} \Delta^6$; the Y -side identity gives $d \mid v_1^{21} \Delta^6$; and $\gcd(u_1, v_1) = 1$ cancels the monomials. $\square$

This is the uniform replacement for what [2] did per curve: for 389a1 an extended Euclidean computation, and for 5077a1 the same plus a Hermite-normal-form argument to show that a stray factor was forced. Both are subsumed. The exponent 6 is the price of the cube in (6); sharpening it for a particular curve remains possible and is not needed downstream, because the content bound is only ever used to know that a content exists and is bounded.

4.5. Size bounds.

Proposition 4.6 (`abs_DDz_le`, `abs_Sfz_le`, `abs_Prz_le`). *With $H_i = \max(|u_i|, |v_i|)$,*

$$|\mathcal{D}| \leq 4H_1^2 H_2^2, \quad |\mathcal{S}| \leq (4 + |b_2| + 2|b_4| + |b_6|)H_1^2 H_2^2, \quad |\mathcal{P}| \leq (1 + |b_4| + 2|b_6| + |b_8|)H_1^2 H_2^2.$$

These are triangle inequalities on bidegree-(2,2) monomials, and they are sharp against the hand computations of [2]: at 5077a1 the $\mathcal{S}$ constant is $4 + 0 + 28 + 25 = 57$ and the $\mathcal{P}$ constant $1 + 14 + 50 + 49 = 114$, which are the numbers derived by hand there. Writing

$$CU_{\text{sec}} := \max(4, 4 + |b_2| + 2|b_4| + |b_6|, 1 + |b_4| + 2|b_6| + |b_8|) = \max(4, CU), \quad (9)$$

all three are bounded by $CU_{\text{sec}} H_1^2 H_2^2$.

There is a matching lower bound, proved from the same certificates in the manner of Proposition 3.2.

Proposition 4.7 (`secant_size_lower`). *There is an explicit $CL_{\text{sec}} = CL^2 \cdot S$, with S the sum of the coefficient-absolute-sums of the eight cofactors of Proposition 4.4, such that*

$$\Delta^4 H_1^2 H_2^2 \leq CL_{\text{sec}} \cdot \max(|\mathcal{D}|, |\mathcal{S}|, |\mathcal{P}|).$$

Remark 4.8. Proposition 4.7 is the honest analogue of Proposition 3.2 on the secant side — two-sided control of the forms themselves — but it is *not* used in the parallelogram assembly below. The lower half of the quasi-parallelogram law comes for free from the doubling trick (§5), and the plan that produced Proposition 4.7 had assumed otherwise. We record the correction rather than quietly presenting the result as load-bearing.

5. THE UNIFORM HEIGHT WINDOW

Lemma 5.1 (`naiveHeight_mul_le_of_quadratic`). *Let $A, B, C \in \mathbb{Z}$ with $A \neq 0$ and let $u, v \in \mathbb{Q}$ satisfy $A(u + v) = B$ and $Auv = C$. Then $H(u)H(v) \leq 2 \max(|A|, |B|, |C|)$.*

Lemma 5.1 is curve-independent and was already available from [2], where it is proved without Gauss's lemma. Combining it with Propositions 4.2 and 4.6 gives the upper half of the window at the point level.

Proposition 5.2 (`secant_upper_window`). *On any Weierstrass curve over $\mathbb{Q}$ with integer b -invariants, for affine points with $x_1 \neq x_2$,*

$$H(x(P+Q)) H(x(P-Q)) \leq 2CU_{\text{sec}} H(x_1)^2 H(x_2)^2.$$

The lower half is not a second estimate but the same one, applied elsewhere. The sum and difference of the pair $(P+Q, P-Q)$ are $2P$ and $2Q$; so Proposition 5.2 at that pair, chained with the duplication lower window of Proposition 3.3, bounds $H(x_1)^2 H(x_2)^2$ from above. Doing so requires knowing $x(P+Q) \neq x(P-Q)$, which is where hypothesis (iii) first bites: equality would force $P+Q = \pm(P-Q)$, hence $2Q = O$ or $2P = O$ (`X_add_ne_X_sub`).

Theorem 5.3 (`point_two_sided`, `lognhX_defect_le`). *Let W have integer b -invariants and $\Delta \neq 0$, and let P, Q be affine points with $x_1 \neq x_2$, neither 2-torsion. Put*

$$C_{\text{par}} := (CL^2 + 1) \cdot 2CU_{\text{sec}}. \quad (10)$$

Then

$$H(x_1)^2 H(x_2)^2 \leq C_{\text{par}} H(x(P+Q)) H(x(P-Q)) \leq C_{\text{par}} \cdot 2CU_{\text{sec}} H(x_1)^2 H(x_2)^2,$$

equivalently $|h(P+Q) + h(P-Q) - 2h(P) - 2h(Q)| \leq \log C_{\text{par}}$.

The $+1$ in (10) is there so that $C_{\text{par}} \geq 2CU_{\text{sec}}$ unconditionally and one constant serves both sides of the window; it costs nothing and removes a case split.

Theorem 5.4 (`heightWindow_universal`). *Let W be admissible. Then $R \mapsto \log H(x(R))$ satisfies the doubling window with constant $\kappa = \max(CU, CL)$ on all of $W(\mathbb{Q})$, with no per-curve input: the integral b -relation is derived from `mathlib`'s, the nonvanishing of Ψ comes from (iii), and the identification of $x(2R)$ with Φ/Ψ comes from `mathlib`'s group law.*

Theorem 5.4 is what makes $\hat{h}$ exist on an admissible curve, by the generic construction recalled at the end of §3. We write $\hat{h} = \text{canheight}W$.

6. THE PARALLELOGRAM LAW AND BI-ADDITIVITY

Theorem 6.1 (`canheight_parallelogram_universal`). *Let W be admissible and let P, Q be affine points with distinct x -coordinates. Then $\hat{h}(P+Q) + \hat{h}(P-Q) = 2\hat{h}(P) + 2\hat{h}(Q)$.*

Proof, as formalized. Hypothesis (iii) forces $2^n R \neq O$ for every n and every $R \neq O$, and forces $x(2^n P) \neq x(2^n Q)$ at every level (else $2^n(P \mp Q) = O$). Theorem 5.3 at $(2^n P, 2^n Q)$, divided by 4^n , bounds the defect of the scaled sequences by $\log C_{\text{par}}/4^n \rightarrow 0$; the scaled sequences converge to the four canonical heights; uniqueness of limits closes it. $\square$

Theorem 6.2 (`par`). *On an admissible curve the parallelogram law holds for every pair $P, Q \in W(\mathbb{Q})$, with no side conditions.*

Proof, as formalized. The degenerate cases $P = O$, $Q = O$, $Q = -P$, $P = Q$ are closed by $\hat{h}(O) = 0$, $\hat{h}(-R) = \hat{h}(R)$ and $\hat{h}(2R) = 4\hat{h}(R)$. Otherwise both points are affine, and equal x -coordinates would force one of the degenerate cases, since two affine points of a Weierstrass curve with the same x -coordinate are equal or negatives (`eq_or_eq_neg_of_x_eq`). So distinct x -coordinates may be assumed and Theorem 6.1 applies. Here $\hat{h}(-R) = \hat{h}(R)$ is hypothesis-free: negation fixes the x -coordinate, so the two defining sequences are literally equal. $\square$

Corollary 6.3 (`mul`). $\hat{h}(kR) = k^2\hat{h}(R)$ for every $k \in \mathbb{Z}$ and every $R \in W(\mathbb{Q})$.

Proof, as formalized. Theorem 6.2 at $((n+2)R, R)$ gives the three-term recurrence $\hat{h}((n+3)R) = 2\hat{h}((n+2)R) + 2\hat{h}(R) - \hat{h}((n+1)R)$; with base values $\hat{h}(R)$ and $\hat{h}(2R) = 4\hat{h}(R)$ a two-step induction gives all $k \in \mathbb{N}$, and $\hat{h}(-R) = \hat{h}(R)$ gives the rest. Because Theorem 6.2 is unconditional the recurrence needs no indexing shift, unlike its conditional counterpart in [2], where the step at $k = 0$ had a torsion difference point and had to be supplied separately. $\square$

Theorem 6.4 (`pairing_add_left`, `pairing_comm`, `biAdditive_pairingW`). *On an admissible curve the pairing $\langle P, Q \rangle = \frac{1}{2}(\hat{h}(P+Q) - \hat{h}(P) - \hat{h}(Q))$ is symmetric and additive in each variable, and $\langle P, P \rangle = \hat{h}(P)$.*

The derivation is Jordan–von Neumann: from $4\langle P, Q \rangle = \hat{h}(P+Q) - \hat{h}(P-Q)$ one takes the four instances of Theorem 6.2 at $(P \pm R, Q)$ and $(Q \pm R, P)$, pairs the mixed differences using $\hat{h}(-u) = \hat{h}(u)$, and adds. Symmetry is immediate from the definition. Because Theorem 6.2 is unconditional, no

side condition has to be propagated through the four instances — which is precisely the bookkeeping that [2] avoided, for two points, by a different route (a hypothesized relation reduces everything to multiples of one point), and paid for, for three points, by proving torsion-freeness.

7. THE RANK PIPELINE

The last step is pure linear algebra and was already curve-independent in [2]: if B is a symmetric bi-additive real form on an abelian group G and $P_1, \dots, P_n \in G$ have $\det(B(P_i, P_j)) \neq 0$, then $n \leq \text{rank}_{\mathbb{Z}} G$ (`rank_ge_of_gram_det_ne_zero`). A relation $\sum v_i P_i = 0$ pairs against each P_j to put v in the kernel of the Gram matrix; a matrix over a field with a nonzero kernel vector has zero determinant; so the map $\mathbb{Z}^n \rightarrow G, v \mapsto \sum v_i P_i$, is injective.

Theorem 1.1 is Theorem 6.4 fed into that statement, and its proof term is two lines. The two specializations (1) and `gram_det_fin_three` are the 2×2 and 3×3 expansions, obtained by `Matrix.det_fin_two / Matrix.det_fin_three` together with $\langle P, P \rangle = \hat{h}(P)$ and symmetry.

It is worth being explicit about what the pipeline consumes and what it does not produce. It consumes: four integers, a proof that they are the b -invariants of the curve, a proof that $\Delta \neq 0$, a proof of no rational 2-torsion, a list of points, and a proof that a determinant is nonzero. It does not produce the last of these. Certifying $\det \neq 0$ for a given curve is an interval-arithmetic computation, done per curve; see §12(c).

8. VALIDATION, AND A DEFINITIONAL COLLAPSE

8.1. The two curves. For $E = 389a1$ the hypothesis package is discharged as follows: the b -invariants $(4, -4, 1, -3)$ by `norm_num` from $(a_1, \dots, a_6) = (0, 1, 1, -2, 0)$; $\Delta = 389 \neq 0$ by `decide` on the integer polynomial; and (iii) from the triviality of $E(\mathbb{Q})_{\text{tors}}$, proved in [2]. For $E' = 5077a1$, with $(a_1, \dots, a_6) = (0, 0, 1, -7, 6)$, the b -invariants are $(0, -14, 25, -49)$ and $\Delta = 5077$, again by `decide`, and (iii) again from triviality of the torsion subgroup. Feeding in the certified regulator determinants of [2] yields (2) and (3). For the three-point case one needs that the 3×3 Gram determinant of the uniform pairing is the determinant computed in [2]; that is `gram_det_eq_regDet3`, closed by `ring` after rewriting, with no slack.

8.2. The definitional collapse. Transporting the numerical certificates required a bridge, because the two layers reach the canonical height through different helper functions: the uniform layer through $\log H$ composed with a generic x -projection, the per-curve layer of [2] through $\log$ of a differently-defined naïve height composed with a per-curve x -projection. The two x -projections agree by `cases` then `rfl` (the pattern matches are the same), and the two naïve heights agree by a cast. This gives the two log-height helpers as *equal functions*.

The point worth recording is what happens next. Once the helpers are equal as functions, the two approximating sequences and hence the two `limUnders` are *definitionally* equal: `canheightW_eq_389` closes by `rfl`, and so does `canheightW_eq_5077`. No analytic argument about limits was needed and none was attempted.

Remark 8.1. We think this is a real point about formalized refactoring rather than an anecdote. The risk in a universalization of this depth is not that a proof breaks — broken proofs announce themselves — but that the object being computed silently changes, so that the new theorems are true of something slightly different from the old ones and the comparison is never actually made. Here the comparison is made, and it is made by the kernel, at the strongest available level: not propositional equality proved by an argument, but definitional equality checked by reduction. The uniform layer and the per-curve layer are the same construction.

9. THE PRICE OF UNIFORMITY

The universal constants are much worse than the hand-built ones. The measured values, for the two curves of [2], are the following.

curve	(b_2, b_4, b_6, b_8)	Δ	CU	$CL = \kappa$	κ from [2]	CU_{sec}	C_{par}
389a1	$(4, -4, 1, -3)$	389	17	672 192	1 728	17	$\approx 1.54 \times 10^{13}$
5077a1	$(0, -14, 25, -49)$	5077	114	536 913 058	105 754	114	$\approx 6.57 \times 10^{19}$

Exactly, $C_{\text{par}} = 15\,362\,630\,885\,410$ for 389a1 and $C_{\text{par}} = 65\,726\,844\,062\,007\,791\,220$ for 5077a1. Every entry is obtained by evaluating the definitions of §§3–5 at the four integers in the second column; the κ of [2] is that paper’s per-curve duplication constant.

Three remarks, in decreasing order of comfort.

The upper side is sharp. $CU = 17$ and $CU = 114$ are exactly the size constants derived by hand in [2] for the two curves. Nothing is lost on the CU side, and hence nothing on CU_{sec} , which by (9) is $\max(4, CU)$.

The loss is entirely in CL , and it is one factor of Δ . The measured ratios are $672\,192/1\,728 = 389$ and $536\,913\,058/105\,754 = 5077$ — in both cases exactly Δ . Neither constant was designed to make this happen. The natural reading is that the universal route normalizes its Bézout identity to the resultant, $F\Phi + G\Psi = \Delta^2 X^7$, and then cancels a content bounded by Δ^2 ; whereas for these two curves the actual content of (Φ, Ψ) is only Δ , and the hand computation of [2] found the sharper identity with Δ on the right. The uniform statement cannot know that, because $\text{Res}(\Phi, \Psi) = \Delta^2$ is what is true in the free polynomial ring. Beyond that structural factor, CL is a sum of absolute values of Bézout cofactors, which is not tight.

It is still the right trade. The cost is visible downstream: the torsion-enumeration argument of [2] searches $\kappa^{1/3}$ -many candidates, so for 389a1 the universal constant would widen the search from 12 to about 88; and $\log C_{\text{par}}$ enters the parallelogram defect, which is squeezed by 4^{-n} , so matching the accuracy that [2]’s constant 10368 gives at 389a1 would need about $\log_4(C_{\text{par}}/10368) \approx 15$ further dyadic levels. Both are real costs, and both are costs in *numerics*, not in *proof*. Against them: a constant that is automatic for every curve, obtained by evaluating a polynomial at four integers, versus a sharp constant that costs a hand-built certificate suite per curve. Sharpening one curve later is optional; deriving a constant at all is not. Nothing above prevents a user from supplying a better per-curve κ to the same generic machinery — that is exactly how [2]’s numerics were produced — and the uniform bound remains as the fallback that always exists.

10. OUTLOOK

Three things now cost less than they did.

A new curve. The scale-out cost is the hypothesis package — four b -invariants, $\Delta \neq 0$, no rational 2-torsion — plus a certified-nonzero Gram determinant at the chosen points. The first three are **decide**-level once the coefficients are fixed; the last is interval arithmetic. Nothing else.

Higher rank. Theorem 1.1 is stated for all n , so $\text{rank} \geq 4$ is a matter of supplying four points and one determinant. The algebra is finished.

Upstreaming. Three pieces are of general interest and are independent of everything else here: Lemma 5.1; the implication “two-sided duplication estimate $\Rightarrow$ canonical height”; and the Gram criterion. A fourth, more speculative candidate is the pair (Proposition 4.3, Proposition 4.4) — the diagonal degeneration and the twin certificates — which is what turns the approximate parallelogram law from a per-curve computation into a theorem with explicit constants. `mathlib`’s `NumberTheory/Height/EllipticCurve.lean` lists the approximate parallelogram law as an open task (§1.4); the route taken here is one way to it, at the cost of specializing the ground field to $\mathbb{Q}$ and buying explicit constants in exchange.

11. VERIFICATION DATA

All results were checked with Lean 4 toolchain `leanprover/lean4:v4.24.0-rc1` against `mathlib` at commit `eed770a434957369c6262aa3fb1d6426419016d4`. At repository commit `b99f2ab6abe7f6df627a4f070122b71882dac52` (5 August 2026), `lake build PF` completes in 4711 jobs with zero errors and zero warnings.

The one-command reproduction is

```
cd PF_Lean4_Code && lake build PF
```

Each source file listed below ends with `#print axioms` declarations for its own results, so the axiom report is printed by that same build and requires no separate step. Every theorem cited in this paper reports a subset of

```
[propext, Classical.choice, Quot.sound]
```

— several of the polynomial identities report only `[propext]` or `[propext, Quot.sound]`. The development contains no `sorry`, no `native_decide` (which would introduce `Lean.ofReduceBool`), and no project-level `axiom` declarations. We do not report a wall-clock figure: the build was replayed from cache for this measurement, so any time we could quote would not be the from-scratch time a reader would see.

The table below maps each claim to its Lean declaration. All files are in `PF_Lean4_Code/PF/`; file names carry an internal revision tag, which is not part of the mathematics.

claim	Lean declaration	file
Prop. 3.1 (Bézout)	<code>bezout_X, bezout_Y</code>	<code>DuplicationBezoutUniversal_r174</code>
Prop. 3.1 (content)	<code>dvd_disc_sq_of_isCoprime</code>	<code>DuplicationBezoutUniversal_r174</code>
Prop. 3.2	<code>max_abs_le, disc_sq_mul_le</code>	<code>DuplicationSizeUniversal_r175</code>
Prop. 3.3 (heights)	<code>nh_dblQ_le,</code> <code>nh_pow_le_of_disc_ne_zero</code>	<code>DuplicationHeightUniversal_r176</code>
Prop. 3.3 (formula)	<code>addX_self_eq_dblQ, psiX_eq_sq</code>	<code>DuplicationFormula_r178</code>
doubling window	<code>heightWindow_of_xdbl_ne_zero</code>	<code>DuplicationLogWindowFixed_r179</code>
generic canonical height	<code>HeightWindow, canheight,</code> <code>canheight_window_shift</code>	<code>CanonicalHeightGeneric_r171</code>
Prop. 4.1	<code>addX_add_addX_negY,</code> <code>addX_mul_addX_negY</code>	<code>SecantUniversal_r181</code>
(8), Prop. 4.2	<code>DDz, Sfz, Prz,</code> <code>Sfz_spec, Prz_spec</code>	<code>SecantSizeUniversal_r193</code>
Prop. 4.6	<code>abs_DDz_le, abs_Sfz_le,</code> <code>abs_Prz_le</code>	<code>SecantSizeUniversal_r193</code>
Prop. 4.3	<code>Sfz_diag, Prz_diag, DDz_diag</code>	<code>SecantContentBridge_r195</code>
Prop. 4.4 (v_2^3 twins)	<code>psi_sq_cert, phi_sq_cert</code>	<code>SecantContentBridge_r195</code>
Prop. 4.4 (u_2^3 twins)	<code>psi_sq_certA, phi_sq_certA</code>	<code>SecantContentUniversal_r196</code>
(6), weight cancellation	<code>dvd_cube_of_bezout,</code> <code>dvd_of_dvd_coprime_weights</code>	<code>SecantContentUniversal_r196</code>
Thm. 4.5	<code>secant_content_universal</code>	<code>SecantContentUniversal_r196</code>
Prop. 4.7	<code>secant_size_lower, CLsec</code>	<code>SecantSizeLower_r197</code>
Lem. 5.1	<code>naiveHeight_mul_le_of_quadratic</code>	<code>QuadraticRootHeights_r148e</code>
Prop. 4.2 (integer form)	<code>sum_relation, prod_relation</code>	<code>SecantUpperWindow_r198</code>
Prop. 5.2, (9)	<code>secant_upper_window, CUseC</code>	<code>SecantUpperWindow_r198</code>
Thm. 5.3, (10)	<code>point_two_sided,</code> <code>lognhX_defect_le, Cpar</code>	<code>PointParallelogramUniversal_r199</code>
$x(P+Q) \neq x(P-Q)$	<code>X_add_ne_X_sub,</code> <code>eq_or_eq_neg_of_x_eq</code>	<code>PointParallelogramUniversal_r199</code>
Thm. 5.4, (7)	<code>heightWindow_universal,</code> <code>b_relation_int</code>	<code>CanheightParallelogramUniversal_r200</code>
Thm. 6.1	<code>canheight_parallelogram_universal</code>	<code>CanheightParallelogramUniversal_r200</code>
Thm. 1.2(5), pairing	<code>pairingW, pairing_self_universal</code>	<code>CanheightParallelogramUniversal_r200</code>

claim	Lean declaration	file
Thm. 6.2, Cor. 6.3	<code>par, mul, canheight_neg,</code> <code>canheight_zero</code>	<code>BiAdditiveUniversal_r201</code>
Thm. 6.4	<code>pairing_add_left, pairing_comm,</code> <code>biAdditive_pairingW</code>	<code>BiAdditiveUniversal_r201</code>
Gram criterion	<code>rank_ge_of_gram_det_ne_zero</code>	<code>GramRankGeneral_r170</code>
Thm. 1.1	<code>rank_ge_universal</code>	<code>RankPipelineUniversal_r202</code>
(1) and $n = 3$	<code>gram_det_fin_two,</code> <code>rank_ge_two_universal,</code> <code>gram_det_fin_three</code>	<code>RankPipelineUniversal_r202</code>
§8, bridges	<code>canheightW_eq_389,</code> <code>canheightW_eq_5077,</code> <code>pairingW_eq_389,</code> <code>gram_det_eq_regDet3</code>	<code>RankTransferWitnesses_r203</code>
(2), (3)	<code>E389a1_rank_ge_two_universal,</code> <code>E5077a1_rank_ge_three_universal</code>	<code>RankTransferWitnesses_r203</code>

The Bézout cofactors of Proposition 3.1, the size constants of Proposition 3.2, and the cofactor size bounds of Proposition 4.7 were produced and re-verified by generator scripts (`codex/gen_r174.py`, `codex/gen_r175.py`, `codex/gen_r197.py`); nothing in them was transcribed by hand. The generators are not trusted: every identity they emit is re-checked in Lean by `ring` or `linear_combination`, and it is the Lean check, not the generator, that this paper cites.

12. WHAT IS NOT PROVED

The value of a formalization is the precision of its claims, so we are explicit.

- (a) **Lower bounds only.** Theorem 1.1 and (2), (3) are rank *lower* bounds. Upper bounds require descent — Selmer groups, the weak Mordell–Weil theorem — which exists in no proof assistant. We make no claim about the exact rank of any curve. That the ranks of 389a1 and 5077a1 are 2 and 3 is known classically [3, 4] and is used nowhere here.
- (b) **No L -functions, no BSD.** Nothing here concerns L -functions, analytic ranks, Tate–Shafarevich groups, or the Birch–Swinnerton-Dyer conjecture. No L -function of an elliptic curve is defined anywhere in the development.
- (c) **The Gram determinant is an input.** Theorem 1.1 takes $\det(\langle P_i, P_j \rangle) \neq 0$ as a hypothesis and does not produce it. Certifying it for a given curve and a given set of points is a separate, per-curve computation — certified rational interval arithmetic on the dyadic chain, as carried out in [2] for the two curves whose certificates are reused here. The pipeline is a machine with an input slot, not an oracle.
- (d) **The 2-torsion hypothesis genuinely restricts the class of curves.** Hypothesis (iii) is not removable by better bookkeeping: Ψ vanishes exactly at the x -coordinates of rational 2-torsion points (§1.3), which is where the duplication map and hence the height window break down. Curves with rational 2-torsion — for instance 102a1, excluded for the same reason in [2] — are not covered by any theorem in this paper. A per-step variant of the window would be needed, and we have not carried it out.
- (e) **Mordell–Weil is neither used nor proved.** All statements are about $\text{rank}_{\mathbb{Z}}$ of the point group as an abstract $\mathbb{Z}$ -module, which is meaningful without finite generation. We do not prove that $W(\mathbb{Q})$ is finitely generated, and no step depends on it.
- (f) **The two curves are validation, not new arithmetic.** (2) and (3) re-derive results first obtained in [2], and their numerical inputs are the certificates of [2]. They are evidence that the uniform machine is non-vacuous and agrees with the hand-built development; they are not new facts about elliptic curves.

- (g) **Optimality is not claimed.** The exponent 6 in Theorem 4.5, and the constants of §9, are what this route yields. We do not claim any of them is best possible, and §9 exhibits a factor of Δ that is known to be lost.

ACKNOWLEDGEMENTS

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